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Influence of Metal Oxide Interface Layers on the Thermochromic Properties of VO_2 Thin Films Deposited by Magnetron Sputtering

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ABSTRACT

VO_2 exhibits thermochromic properties with a metal-to-insulator transition occurring at 68°C. VO_2 thin films were deposited by RF magnetron sputtering on Al-doped ZnO coated fused silica with Al_2O_3, TiO_2, and WO_3 as interface layers with the aim of controlling the transition temperature. Annealing under nitrogen atmosphere was employed to promote diffusion or doping from the interface layer into the VO_2 films. The results show that matching between crystalline structures and high oxygen flows during deposition resulted in a higher degree of doping or diffusion of the interface layer into the VO_2. Consequently, oxygen flow rate tuning during the VO_2 deposition on top of a WO_3 interface layer, followed by annealing enabled a decrease of the metal-to-insulator temperature from 68°C to 55°C.

1 | Introduction

Buildings are responsible for the consumption of around 30% of the world's primary energy, hence, the need to decrease the power consumption utilized for cooling, heating and illumination by controlling the energy transfer through windows [1–4] For this purpose, different types of windows have been proposed [5–7] where a promising solution would be the use of functional metal oxide thin films with thermochromic properties based on VO_2 [8–11]. VO_2 is insulator at room temperature while upon heating, a sharp change occurs at 68°C toward a metallic phase that exhibits high reflection of the infrared light and a much lower resistivity [12]. This metal-to-insulator transition (MIT) has made VO_2 thin films appealing for various applications, including batteries [13, 14], bolometers [15–18], optoelectronic devices [19–22], and smart windows [23–25], among others. VO_2 can be deposited by atomic layer deposition [26–30], chemical vapor deposition [31–33], evaporation [34–36], molecular beam epitaxy [37–39], pulsed-laser deposition (PLD) [40–43], sol-gel deposition [44–46], and magnetron sputtering [47–49]. Among all, magnetron sputtering provides very good compatibility with semiconductor processes, high uniformity over large area, good adhesion, and high crystal quality [50, 51]. When magnetron sputtered, VO_2 thin films can obtained by reactive-sputtering from vanadium [52–54], or from oxide targets such as V_2O_3 [55, 56], VO_2 [57, 58], and V_2O_5 [59–61]. This versatility results from the ability of vanadium to form a variety of oxide compounds, each of them presenting distinct properties and implications. However, this also makes the deposition of pure VO_2 extremely challenging [62].

In this context, besides the fine tuning challenge of the deposition process to mainly obtain VO_2, the desired use of VO_2 for smart windows requires to reduce the MIT temperature [63, 64]. So far, the epitaxial deposition of VO_2 by PLD on an Al_2O_3 substrate with a TiO_2 buffer layer has achieved a substantial strained VO_2, which exhibited a lower MIT temperature of 44°C [65]. Similar results were obtained for VO_2 epitaxially grown on a TiO_2 substrate using RuO_2 as buffer layer [66]. Typical defects in metal oxide thin films

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TABLE 1 | Sputtering parameters used in the deposition of metal oxide thin films.

Parameters	AZO	Al_2O_3	TiO_2	WO_3	VO_2
Pressure (mTorr)	2	3	3	5	3
Power (W)	140	350	350	140	100
O_2 flow rate (sccm)	0	7	5	5	0.2–0.8
Ar flow rate (sccm)	50	50	50	40	50
Temperature (°C)	200	400	400	400	450
Deposition time (s)	2250	600	600	900	2000–8000

are oxygen vacancies, and VO_2 is not an exception. It was found that the presence of oxygen vacancies in VO_2 could promote the decrease of the MIT temperature [67]. The simulation of VO_{2-x} showed that the transition temperature could be reduced down to −131°C by the formation of additional oxygen vacancies, resulting in a higher number of free electrons [68]. Doping VO_2 with different elements can affect the carrier concentration and introduce structural distortions. Both can significantly affect the temperature at which the transition takes place [69, 70]. For example, it was recently reported that annealing of VO_2 deposited on AZO promoted Zn diffusion accompanied by an increased MIT [71]. The incorporation of W atoms has also proved effective in reducing the MIT temperature. For example, W-doped VO_2 has been fabricated by solution combustion method that enabled the decrease of the transition temperature down to 15°C [72]. Consequently, so far, the decrease of the MIT temperature has been achieved by precise deposition or doping, carried out using complex processes.

The aim of this study is to investigate the influence of the oxygen flow and annealing on the thermochromic properties of VO_2 deposited on AZO precoated fused silica with Al_2O_3, TiO_2, and WO_3 as interface layers for reducing MIT.

2 | Methods

AZO thin films were deposited on 4-inch fused silica wafers using a Kurt Lesker 2017 PRO Line PVD75 thin film deposition cluster system from a 3-inch AZO target containing 2 wt. % of Al. The AZO precoated fused silica wafers were subsequently used as substrates either to deposit VO_2, or one of the different interface layers studied. Al_2O_3, TiO_2, and WO_3 thin films were deposited from a 3-inch Al target, Ti target, and WO_3 target provided by Kurt Lesker. Reactive-pulsed DC magnetron sputtering using 100 kHz and a reverse time of 1 µs was employed to deposit Al_2O_3 and TiO_2. High pulse impulse magnetron sputtering (HiPIMS) [73] was utilized to deposit WO_3 using a voltage of 900 V, and a pulse length of 40 µs. Then, the precoated wafers were cut into 2 × 1.5 cm^2 samples and ultrasonically cleaned. VO_2 thin films were deposited on top of one of the interface layers utilizing RF magnetron sputtering from a 3-inch VO_2 target provided by Testbourne, Scientific Materials and Instruments. The main sputtering parameters chosen for each deposition are described in Table 1.

To identify the crystalline phases, X-ray diffraction (XRD) spectrum θ-2θ from 10° to 50° using a 5 mm-length Soller slit, a step of 0.01°, and a speed of 0.5°/min was recorded using a Rigaku Smartlab X-ray diffraction system.

A K-alpha x-ray photoelectron spectrometer (Thermo Fisher Scientific) was used to study the elemental composition of films. A monochromatic Al-Kα x-ray source with a spot size of 400 µm was used. The survey spectrum was obtained from 0 eV to 1350 eV, using a pass energy of 200 eV, 5 scans, and a dwell time of 50 ms. The Zn_{2P}, O_{1s}, Al_{2P}, C_{1s}, V_{2p}, Ti_{2p}, and W_{4f} peaks were obtained using a pass energy of 50 eV, 5 number of scans and a dwell time of 50 ms. The C_{1s} peak was used as a reference for 284.8 eV. The Thermo Fischer Avantage Software was used to determine the composition and peak fitting using 100 maximum iterations of a product of a Gauss-Lorentz mix to fit all the levels. An Ar ion gun was used to measure the depth profile using 1000 eV of ion energy for 20 s between the 10 etched levels.

The thickness was measured using an Ellipsometer VASE from J.A. Woollam. Optical transmittance was obtained using a UV–vis-NIR CRAIC 20/30 PV Microspectrophotometer with an incorporated Linkam PE120 heating stage to record the transmittance from 500 nm to 2100 nm wavelength as a function of temperature.

3 | Results and Discussion

3.1 | VO_2 Thickness Effect

The first investigation consisted in measuring the optical transmittance as a function of temperature for VO_2 thin films deposited on AZO/TiO_2 (110 nm AZO and 67 nm TiO_2) and AZO/WO_3 (110 nm AZO and 96.25 nm WO_3) precoated fused silica using different deposition time, from 2000 to 8000 s leading to thicknesses ranging from 30 to 110 nm. The oxygen flow and deposition temperature used were 0.4 sccm and 450°C, respectively. The corresponding temperature-dependent transmittance and XRD spectra are shown in Figure 1.

Interestingly, the measurements revealed an MIT of VO_2 films without respect to the VO_2 film thickness (50–200 nm) when deposited with a TiO_2 interface layer Figure 1a while only the VO_2 of 50 nm exhibited the MIT when deposited with an WO_3 interface layer (see Figure 1b). All the XRD spectrum exhibited a high peak at around 34.5° of 2θ that corresponds to the AZO *c*-axis oriented [74]. The films thinner or thicker than about 50 nm deposited with the WO_3 interface layer failed to exhibit thermochromic properties due to the lack of crystallinity, or the formation of other phases in detriment of VO_2 (020) or (011)-oriented films, respectively. The use of a TiO_2 (004)-oriented seed layer promoted the formation of VO_2 (020)-oriented films as it is observed in Figure 1c, which showed thermochromic properties independently of the film thickness (deposition time) as presented in Figure 1a. Both, VO_2 and TiO_2, possess a very similar structure, which allows a fiber-textured growth of VO_2 thin films on top of TiO_2 [75]. However, the VO_2 deposited for 2000 s (52 nm in thickness) showed a higher MIT temperature than the other VO_2/TiO_2 stack of layers due to stress, or a rather diffused interface, which led to the merging of the two XRD peaks as observed in Figure 1d. Concerning the dominant intense

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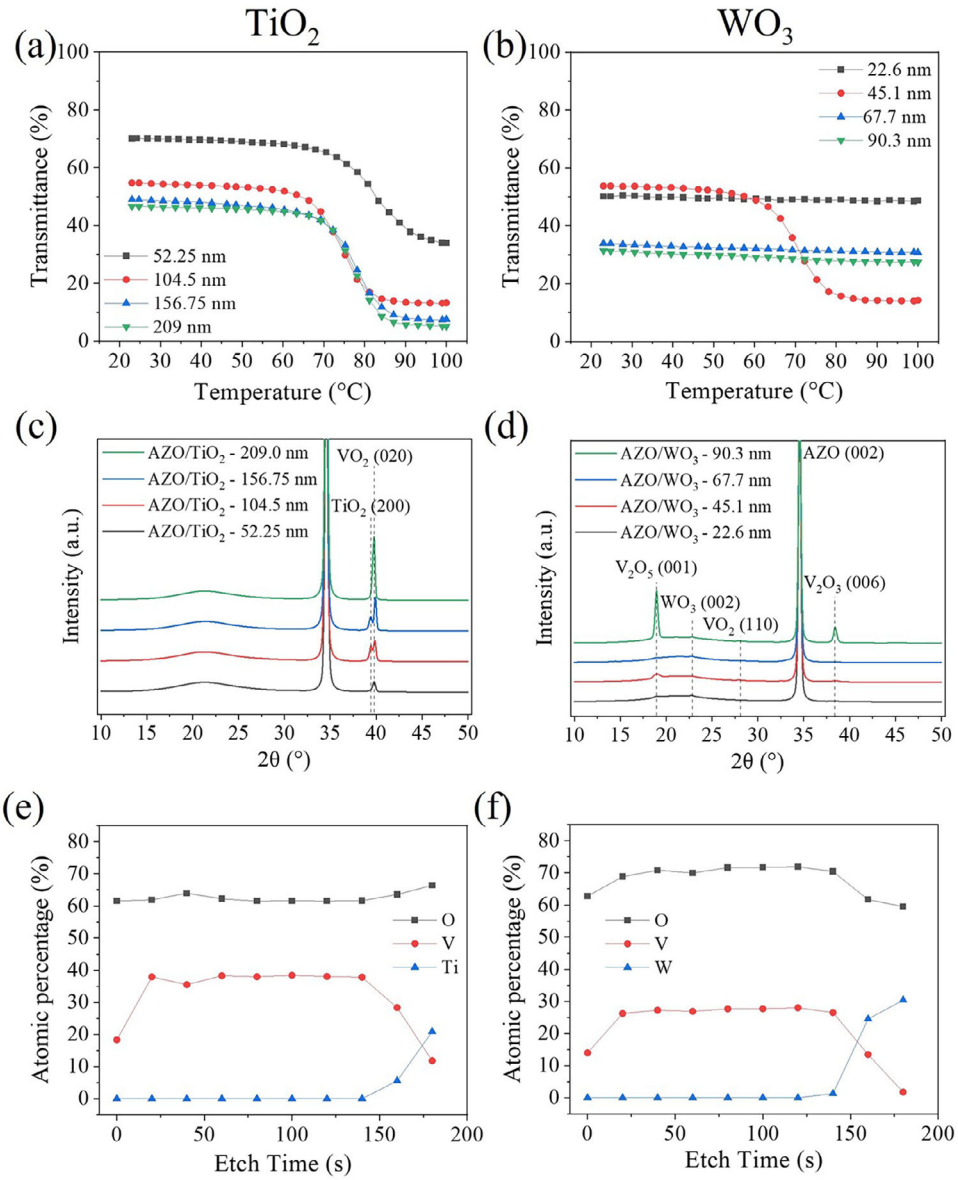


FIGURE 1 | Influence of the film thickness on the MIT of VO_2 deposited on (a) TiO_2 and (b) WO_3, and XRD spectra of vanadium oxide films deposited on (c) TiO_2, and (d) WO_3, and the depth profile of the atomic percentage for (e) TiO_2 and (f) WO_3.

peak of AZO (Wurtzite, hexagonal structure) compared to WO_3 (monoclinic) and TiO_2 (rutile) it is important to mention that the XRD peaks of cubic and hexagonal structures are usually of higher intensity than the structures not presenting many planes fulfilling Bragg's law. Furthermore, AZO is highly textured showing a preferential orientation, which produces higher peak intensities. TiO_2 and WO_3 are polycrystalline with random orientations distributed among many planes. The intensity of each (hkl) reflection also depends on the structure factor (F_{hkl}), which is affected by the atomic scattering factor, and both, Zn and Al have high atomic scattering power. Even if the films are crystalline, the structural factor values for the reflections measured, can differ by orders of magnitude depending on crystal structure and atomic composition [76]. In this case, AZO presents a highly textured *c*-axis oriented peak from a hexagonal wurtzite structure, while monoclinic, rutile structures are more complex presenting more defects or different orientations that can decrease by orders of magnitude the signal measured. This means that although the thicknesses of the AZO, TiO_2, and WO_3 layers were comparable, the XRD patterns differ significantly.

3.2 | O_2 Flow Rate Influence

Once the optimal film thickness for the interface layer was determined, the investigation of O_2 flow rate on VO_2 was carried out using three different interface layers (Al_2O_3, TiO_2, and WO_3). The as deposited Al_2O_3 interface layer was amorphous, with no remarkable changes in XRD, XPS, or VO_2 optical properties. Consequently, it was excluded from Figure 1. However, it was important to assess the VO_2/Al_2O_3/AZO stack from the point of view of Al_2O_3 acting as a capping layer or to capture if Al diffusion into the VO_2 layer could induce optical properties changes.

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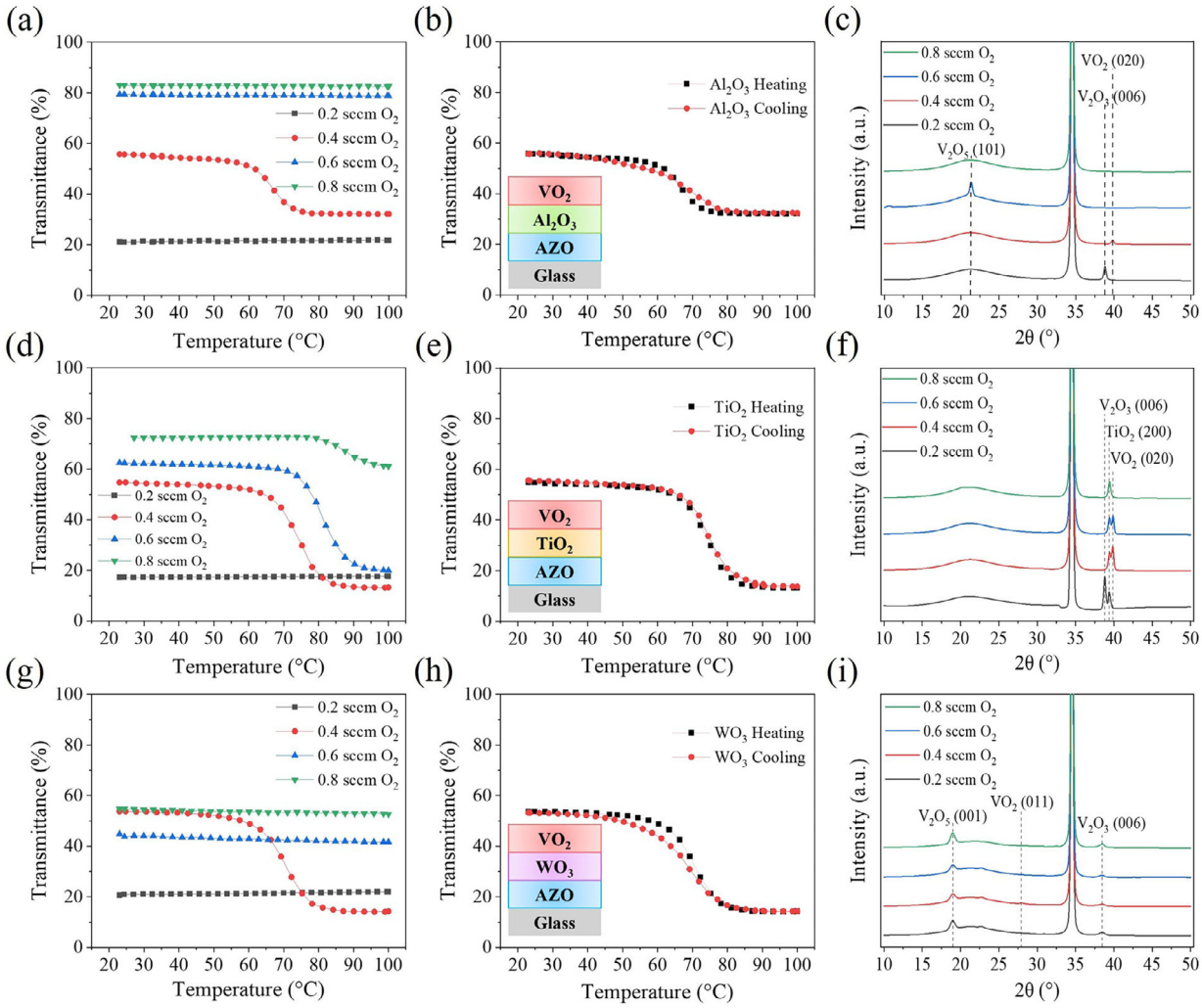


FIGURE 2 | Influence of the oxygen flow rate on the as-deposited vanadium oxide films on the MIT (first column), MIT hysteresis loop (second column), and XRD spectrum (third column), (a–c) using an Al_2O_3 interface layer, (d–f) a TiO_2 interface layer, and (g–i) a WO_3 interface layer.

The temperature-dependent transmittance, the hysteresis loop of the thermochromic samples, and the XRD spectrum of all the stacks of layers as-deposited, including the use of an Al_2O_3 interface layer, are presented in Figure 2.

The use of 0.4 sccm O_2 mixed with 50 sccm Ar promoted the formation of thermochromic films, regardless of the substrate used. These films consisted of VO_2 (020)-oriented or (011)-oriented films as shown in Figure 2c,f,i. The use of higher O_2 flows revealed a higher degree of oxidation forming V_2O_5 (011)-oriented or amorphous films that did not exhibit an MIT in the stack of layers made of AZO/Al_2O_3/VO_2, and AZO/WO_3/VO_2 (see Figure 2a,d,g), respectively. However, the use of TiO_2 interface layers favorized the deposition of thermochromic VO_2 (020)-oriented films, as presented in Figure 2f due to strong match between the rutile (200) TiO_2 and the monoclinic structure of (020)-oriented VO_2 films. The very broad peak spanning from 15° up to 25° present for all samples is the reflection from the fused silica substrate.

These results indicate that the growth mechanisms and the resulting structural properties of vanadium oxide are, as expected, highly dependent on the oxygen flow rate used during the magnetron sputtering deposition.

For a better understanding of the role played by the interface layers, XPS depth profiles were carried out for each structure until the layers placed below the vanadium oxide thin films were reached. The deconvolution of peaks, and analysis of each oxidation state was carried out after each etching step, leading to a depth profile of each stack as it is presented in Figure 3.

Figure 3 shows that no diffusion from the interface layer is taking place. The samples deposited using low oxygen flow shown in the first column tend to be thicker than the samples deposited using a higher oxygen flow and presented in the last column (see (c), (f), and (i)). None of the samples presented a unique pure phase, but rather showing different oxidation states coexisting in the same vanadium oxide thin film. Only VO_2 thin films are supposed to possess a thermochromic effect between room temperature and 80°C, therefore, XPS is a useful method to characterize these films, but caution must be taken, and other characterization methods should be employed along with it. A higher amount of V_2O_5 was detected in the films deposited at higher oxygen

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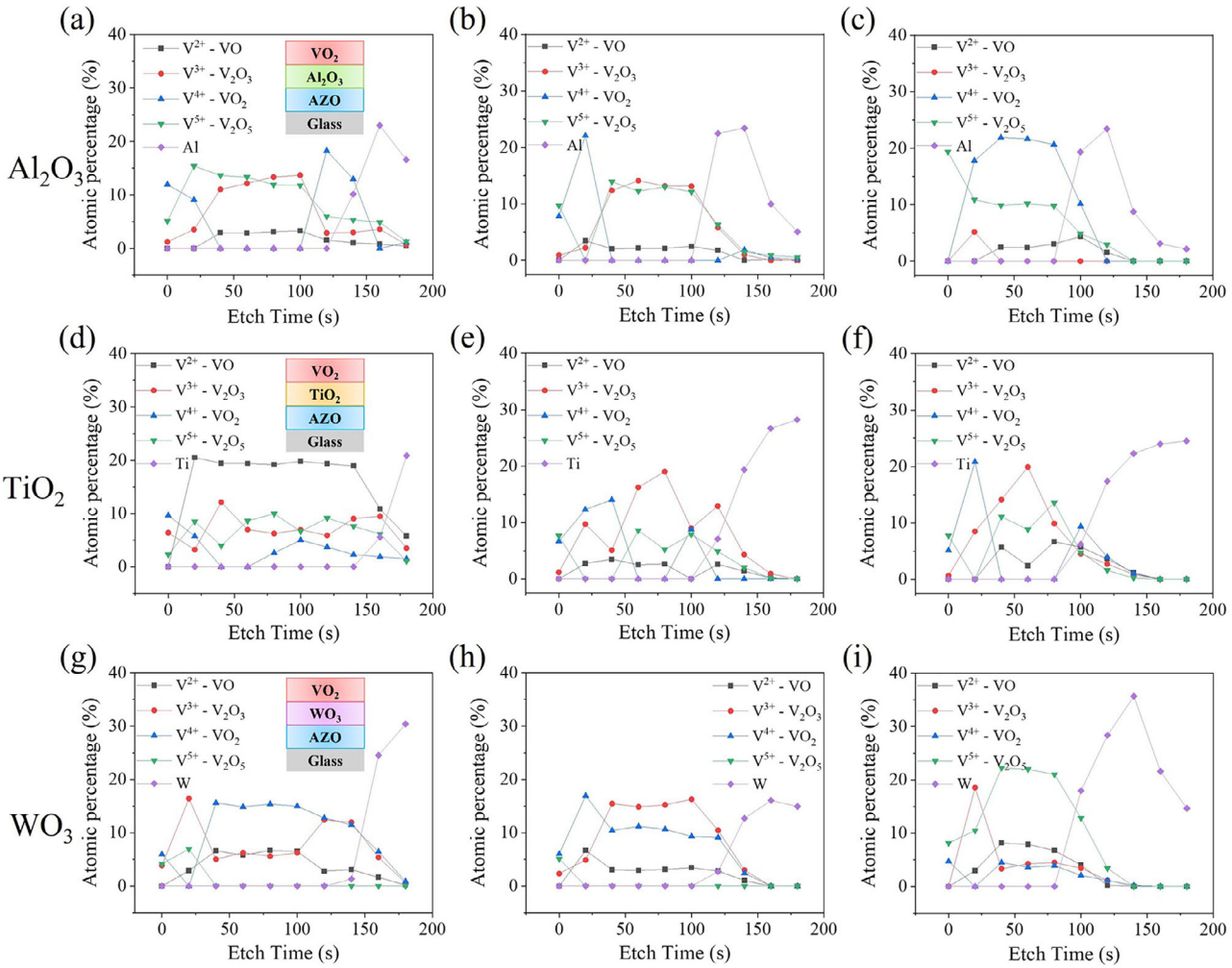


FIGURE 3 | Depth profile of elemental composition of vanadium oxide thin films measured using XPS using 0.4 sccm O_2 (first column), 0.6 sccm O_2 (second column), and 0.8 sccm O_2 (third column) using (a–c) Al_2O_3 interface layer, (d–f) TiO_2 interface layer, and (g–i) WO_3 interface layer.

flow. However, V^{4+}, which corresponds to VO_2, was observed at some etching level for all films. The V_{2p} high resolution XPS patterns deconvoluted to study the oxidation states present in the vanadium oxide thin films are presented as example in Figure S1 when using oxygen flows of 0.5 and 1 sccm.

The choice of interface layers was determined by seeking transparent metal oxides with different oxidation states (Al^{3+}, Ti^{4+}, W^{6+}) that could diffuse into the VO_2 thin films and consequently influence the thermochromic properties. To study this effect, an annealing under N_2 atmosphere was carried out at 500°C for 1 hour. The temperature-dependent transmittance, XRD spectrum, and XPS depth profiles as a function of the oxygen flow during the sputtering deposition are represented in the following figures.

3.3 | AZO/Al_2O_3/VO_2 Annealed Stack

An Al_2O_3 seed layer was deposited due to the Al^{3+} oxidation state, and the ability of Al_2O_3 to retain the Zn from diffusing into the VO_2 thin layer [71]. Their temperature-dependent transmittance, XRD spectra, and XPS depth profiles were recorded as a function of the oxygen flow rate used during the sputtering deposition are shown in Figure 4.

The use of an Al_2O_3 seed layer promoted the formation of thermochromic layers upon annealing for a wider range of oxygen flows during the deposition as compared to using a bare AZO layer, as is observed in Figure 4a. The XRD spectra, shown in Figure 4b, as a function of oxygen flow rate used during the sputtering process revealed the formation of V_2O_3 (006)-oriented films after annealing when the oxygen flow rate utilized was 0.2 sccm. If the oxygen flow rate was increased to 0.4 sccm, the annealing provided a VO_2 (020)-oriented film showing reversible thermochromic properties. Further increase of the O_2 flow rate, formed films that exhibited MIT transition. These films showed a difference in transmittance after heating and before cooling shown at their edges, indicating some kind of irreversibility. The XPS spectrum of the vanadium oxide compounds revealed a high diffusion of Al which can explain the suppression of the (011) V_2O_5 peak found by the as-deposited sample when an O_2 flow rate of 0.6 sccm was found. The use of a high O_2 flow of 0.8 sccm provided an amorphous film that did not crystallize upon annealing and did not show a high diffusion of Al within the vanadium oxide

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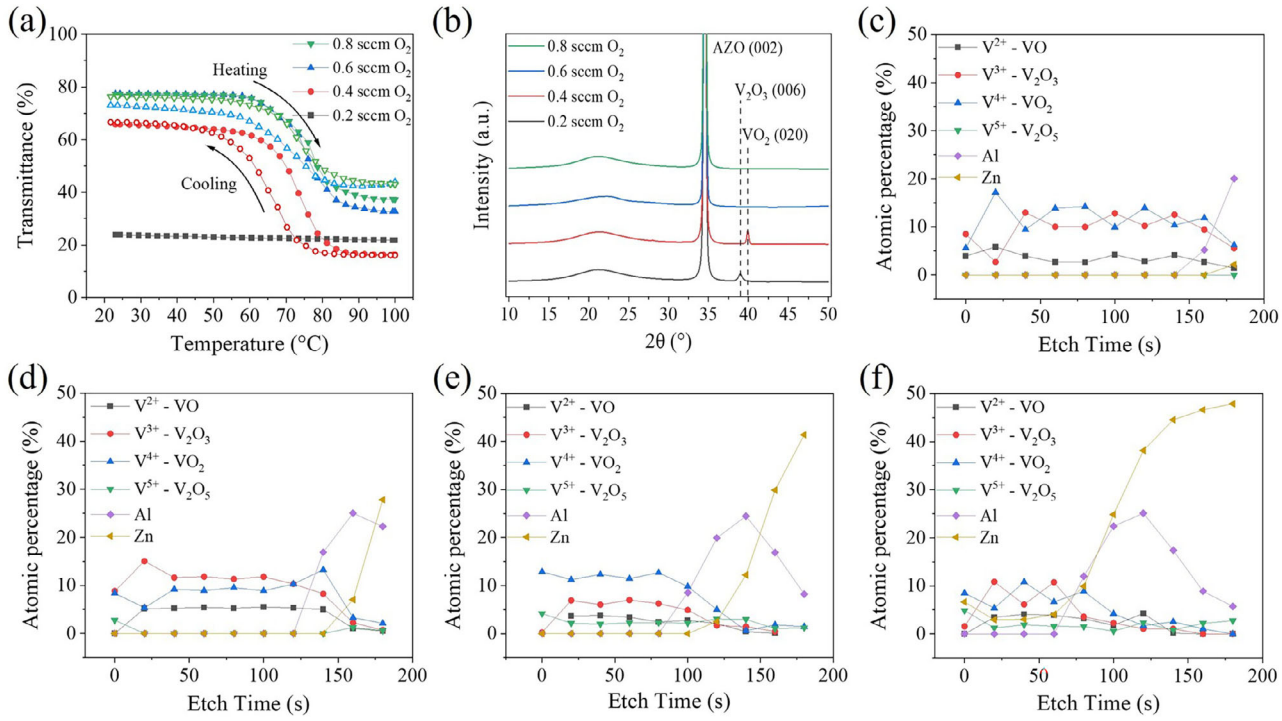


FIGURE 4 | (a) Temperature-dependent transmittance of annealed vanadium oxide thin films deposited on Al_2O_3 seed layer using different oxygen flow rates, (b) Influence of the O_2 flow on the XRD spectra of annealed films, and depth profile of elemental composition of the stack of layers deposited using an O_2 flow rate of (c) 0.2, (d) 0.4, (e) 0.6, and (f) 0.8 sccm.

amorphous layer. Although an MIT temperature was found for the films deposited at O_2 flow rates higher than 0.2 sccm, the derivatives of the heating curves suggest that there is an increase in the MIT temperature to 73.4°C, 76°C, and 78.6°C, when the O_2 flow rate used corresponded to 0.4, 0.6, and 0.8 sccm, respectively.

3.4 | AZO/TiO_2/VO_2 Annealed Stack

To further compare the effect of the oxidation state of the seed layer, Ti^{4+} was used in its oxidized state (TiO_2). When the TiO_2/VO_2 stack of layers was annealed, the loss of thermochromic properties and the transition temperature was observed independently of the oxygen flow rate used, as is seen in Figure 5a.

The thin films forming the TiO_2/VO_2 stack of layers followed the usual trend, showing low values of transmittance through all the range of temperatures of films that were deposited using low oxygen flow. The use of 0.4 sccm of oxygen provided irreversible thermochromic properties as it can be seen in the transmittance difference between the heating and cooling curve at room temperature. If the first derivative of the heating curve is measured, the resulting minimum of the first derivate of the 0.4 sccm, and 0.6 sccm curve are found at 31.9°C, and 60.25°C, respectively. However, the results showed a darkening of the stack deposited using 0.4 sccm of O_2 flow rate at room temperature after cooling, and a small difference in transmittance between states using higher O_2 flow rate. The XRD spectrum shown in Figure 5b revealed different peaks from the metal oxide thin films consisting of V_2O_3 (006)-oriented using 0.2 sccm of O_2 flow, TiO_2 (200)-oriented, and VO_2 (020) phase using 0.4 sccm of O_2 flow rate. The annealed stack of films that were deposited at higher oxygen flows than 0.4 sccm of O_2, showed an amorphous structure with no evident crystalline phase. Due to limited resolution, it is difficult to assess if the peaks associated with VO_2, TiO_2 and V_2O_3 are shifting or simply appearing and disappearing. The XPS patterns revealed low diffusion when low oxygen flow was used as presented in Figure 5c, and an increase in diffusion of the Ti atom within the vanadium oxide films as the oxygen flow was increased, which could lead to the formation of an intermediate metal oxide containing similar amounts of vanadium and titanium as seen in Figure 5d,e. Since Ti has an atomic radii smaller than the vanadium atom [77], it is believed that Ti atom can diffuse large distances and in higher amounts within the vanadium oxide compound during the annealing. The high diffusion of Ti decreases the thermochromic performance of VO_2, proving that TiO_2 is a poor choice as a seed layer for VO_2 thermochromic properties. However, if the stack of layers is not annealed, TiO_2 can stabilize the formation of VO_2 (020)-oriented even if large amounts of O_2 are used, as it was observed in Figure 5f.

3.5 | AZO/WO_3/VO_2 Annealed Stack

Finally, the use of a WO_3 seed layer deposited on top of an AZO electrode layer was studied due to the great success of W^{6+} being a dopant of VO_2 achieving a decrease in its MIT temperature [78]. The deposition of W-doped VO_2 thin films is usually carried out by reactive co-sputtering using metallic targets [79], or using another technique with significantly less precursor of W than the one that provides the V content [80]. In our case, we wanted to study if W^{6+} ions or W metallic atoms coming from the WO_3

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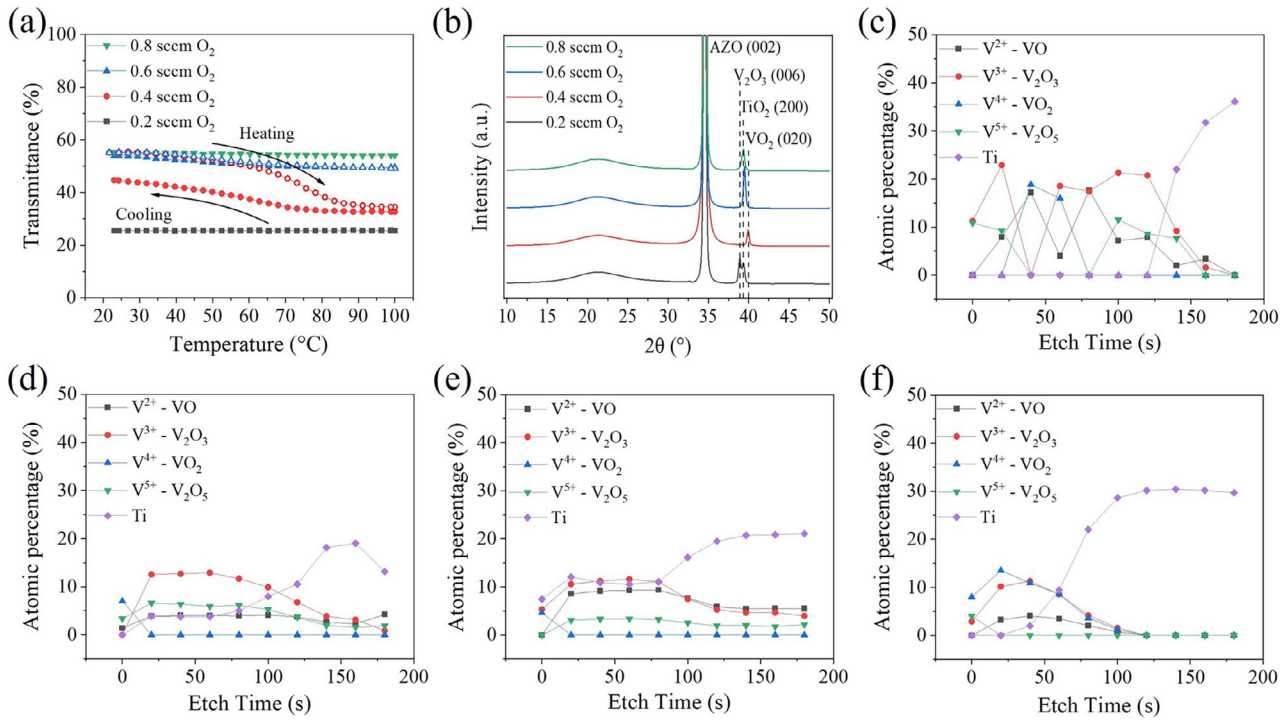


FIGURE 5 | (a) Temperature-dependent transmittance of annealed vanadium oxide thin films deposited on TiO_2 seed layer using different oxygen flow rates, (b) Influence of the O_2 flow rate on the XRD spectra of annealed thin films, and depth profile of elemental composition of the stack of layers deposited using an O_2 flow rate of (c) 0.2, (d) 0., (e) 0.6, and (f) 0.8 sccm.

seed layer could diffuse within the VO_2 thin film, thus decreasing the MIT temperature. The results obtained after annealing of the stack of layers formed by AZO/WO_3/VO_2 are presented in Figure 6.

The hysteresis loop of the thermochromic stack of layers made by WO_3/VO_2 using an O_2 flow rate between 0.4 and 0.8 sccm is represented in Figure 6a. These results indicate that a low O_2 flow of 0.2 sccm was not enough to promote the formation of thermochromic VO_2. However, as the oxygen flow was increased from 0.4 to 0.6 sccm, and to 0.8 sccm, the MIT temperature of the films varied, and its first derivative minimum, or MIT temperature, was found at 68°C, 55°C, and 60°C while heating the stack of layers, respectively. If the minimum of the first derivative was measured upon cooling, the sample deposited at 0.4 sccm of O_2 presented the same minimum at 68°C, but further increase revealed a transition found at 47°C using 0.6 sccm, and at 40°C using 0.8 sccm of O_2 flow, thus indicating a bigger hysteresis loop as the oxygen content of the samples increased while sputtering. The XRD spectrum in Figure 6b revealed a weak VO_2 (011)-oriented film if the optimized oxygen flow rate was used. In all films, V_2O_5 (001)-oriented and V_2O_3 (006)-oriented were found, indicating that the use of WO_3 seed layer influenced the crystalline phase of vanadium oxide thin films after annealing. Moreover, little diffusion of W atoms was detected within the vanadium oxide layer, as observed in Figure 6c–f, which indicates that the grains responsible for the decrease in the MIT temperature are the ones present at the interface.

3.6 | Energy-Saving Performance

Following the smart windows application aim of this study it is important to assess the energy-saving performance of the stacks. Considering the reduction in transmitted solar power as $\Delta Q = \Delta T_{sol} \cdot G$, where G is the incident irradiance, the term ΔT_{sol}, that corresponds to the solar modulation ability, obtained from the optical spectra, directly quantifies the energy-saving potential per unit irradiance and allows a comparison with previous VO_2 smart-window studies. Being ΔT_{sol} measured as indicated below [81].

$$\Delta T_{sol} = T_{sol,cold} - T_{sol,hot} \quad (1)$$

The measured solar modulation ability is reported in Table S1 for the as deposited samples and in Table S2 after annealing. The differences are minimal between all layers except for the VO_2 grown on Al_2O_3 that due to its amorphous character might have reduced the solar modulation ability of the VO_2 grown on top of it. A comparison of the results presented with representative studies is included in Table 2.

The VO_2 thin films presented in this work exhibit a notable energy-saving performance while maintaining high visible transparency, making them well suited for smart-window applications. Our results indicate that the solar modulation ability decreases with decreasing phase-transition temperature, which can be attributed to the flattening of the temperature-dependent transmittance curve. Nevertheless, this study demonstrates that

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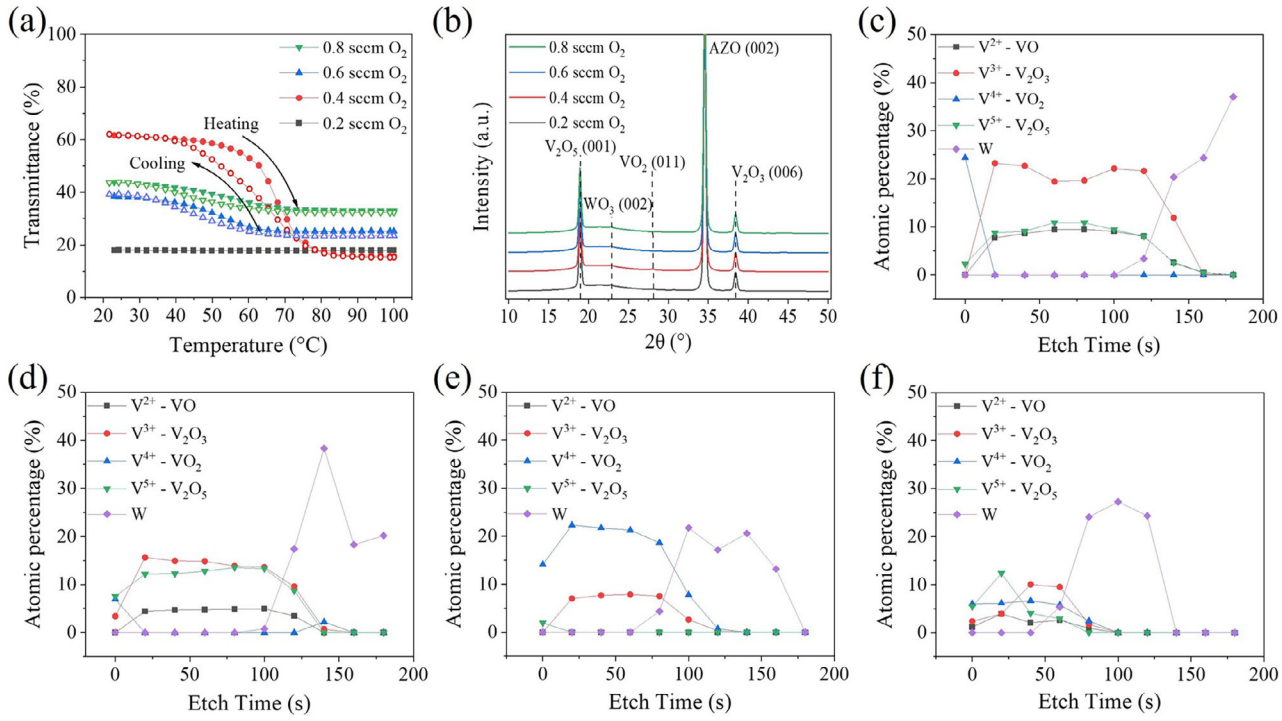


FIGURE 6 | (a) Temperature-dependent transmittance of annealed vanadium oxide thin films deposited on WO_3 seed layer using different oxygen flow rates, (b) Influence of the O_2 flow on the XRD spectra of annealed thin films, and depth profile of elemental composition of the stack of layers deposited using an O_2 flow rate of (c) 0.2, (d) 0.4, (e) 0.6, and (f) 0.8 sccm.

TABLE 2 | Comparison of energy/saving parameters of VO_2 samples.

Material or Stack	$T_{lum,cold}$ [%]	ΔT_{sol} [%]	Refs.
VO_2	23.9	21.0	[81]
AZO/PDLC/VO_2	39.9	9.95	[82]
Nanoporous VO_2	43.3	14.1	[83]
Zn-doped VO_2	41.3	15.3	[84]
W-doped VO_2	61.7	11.7	[85]
WO_3/VO_2 (0.4 sccm O_2 - annealed)	44.6	9.5	This work

an optimized combination of sputtering and post-deposition annealing is an effective approach for fabricating high-quality VO_2-based smart windows.

4 | Conclusions

The study of the influence of Al_2O_3, TiO_2, and WO_3 used as interface layers on the thermochromic properties of vanadium oxide thin films sputtered from a VO_2 target was carried out. Independently of the substrate used, a certain amount of oxygen gas is required to obtain thermochromic VO_2 and to avoid the formation of sub stoichiometric VO_2. All the stacks revealed thermochromic properties as deposited if the oxygen flow rate was optimized. After annealing, each stack was carefully characterized again using XRD, XPS depth profiles, and temperature-dependent transmittance measurements. The AZO/Al_2O_3/VO_2 stack showed a significant recovery of thermochromic properties. except for the one deposited using low oxygen flow rate, suggesting that Al^{3+} doping can benefit VO_2 thermochromic properties. A high diffusion of Ti in the AZO/TiO_2/VO_2 stack was observed, leading to a loss of VO_2 thermochromic properties. Lastly, a small diffusion of W was detected in the VO_2 thin films after annealing the AZO/WO_3/VO_2 stack. However, the VO_2 thin films deposited using 0.6 sccm of O_2 flow rate exhibited a decrease of the MIT temperature from 68°C to 55°C with an optical difference of 15% after annealing. These results indicate that the MIT is highly influenced by the grains formed at the interface between the layers and the VO_2 thin films, and that interface layers can either enhance or degrade performance depending on the choice of materials. Although the decrease in MIT temperature of W-doped VO_2 has been previously reported, this is, to the best of our knowledge, the first time when a WO_3 interface layer promoted this reduction in the MIT temperature. This study opens new directions for the formation and characterization of multilayer stacks made of metal oxide thin films.

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Author Contributions

Eduard Llorens Balada contributed to methodology (equal), investigation (lead), formal analysis (lead), conceptualization (equal), visualization (lead), writing – original draft (equal). Eugen Stamate contributed to writing – original draft (equal), conceptualization (equal), methodology (equal), supervision (lead).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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